

Life cycles

DIFFERENT PLANTS AND ANIMALS GROW TO MATURITY IN DIFFERENT WAYS.

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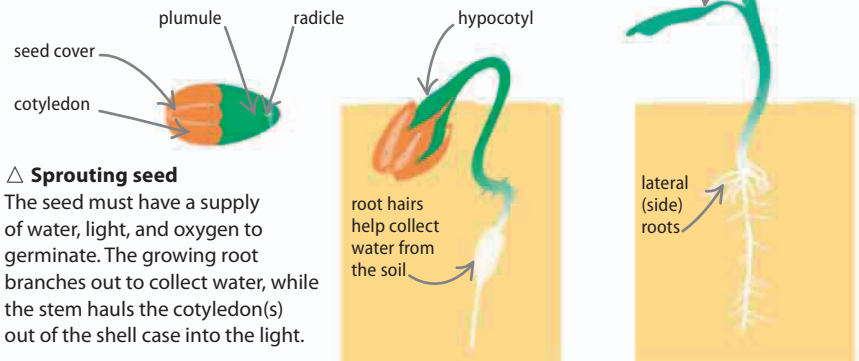
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The early, or juvenile, phase of a multicellular organism's life is devoted to growth. Organisms use a range of systems to reach an adult size, only then developing sexual organs and reproducing.

Germination

A seed is a plant embryo. It already has a root (radicle) and a tiny stem (plumule) inside. The embryonic leaf, called a cotyledon, is a food store that powers the first stage of growth, known as germination. Germination is stimulated by environmental conditions. Longer days—indicating the approach of spring—are a common cue. Some seeds require other cues, such as temperature changes, being soaked in water for a long period, or even heat from fire.



△ Sprouting seed

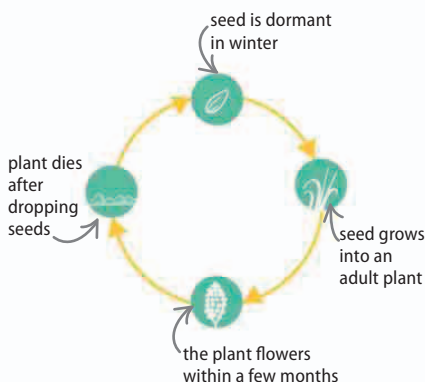
The seed must have a supply of water, light, and oxygen to germinate. The growing root branches out to collect water, while the stem hauls the cotyledon(s) out of the shell case into the light.

Plant life cycles

All flowering plants produce seeds but they do it according to one of three life cycles. Annual plants, such as grasses, sprout, seed, and then die all within one year. Biennials spend the first year growing a storage root, such as a carrot, which then resprouts and flowers in the second year. Perennials live for more than two years and produce repeated batches of seeds.

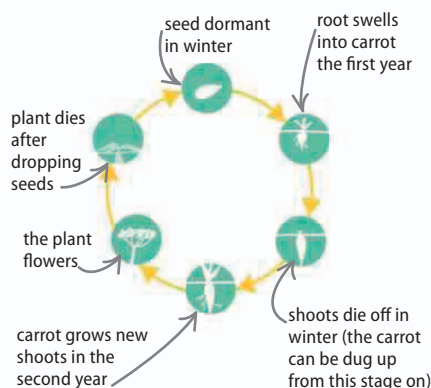
▽ Annual (grass)

The grass seed stays in the soil during winter, grows rapidly, and flowers within a few months. The plant drops new seeds onto the fresh soil before it dies.



▽ Biennial (carrot)

In the first year, leaves above ground fuel the creation of a carrot root, which remains even when the leaves and shoots die off over winter. The next spring, the carrot root's stored sugar fuels new shoots, rapid flowering, and seed production.



▽ Perennial (oak tree)

The oak tree grows for several years before flowering for the first time. Its seeds are dispersed by animals. During winters, the plant becomes dormant, before growing more and flowering again the following year.

